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Magnetic measurement discrepancies in the Ionosphere prior to the 2025 Kamchatka M_w 8.8 earthquake

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Abstract

Magnetic field data from Swarm's magnetometers were investigated during the 2025 M_w 8.8 Kamchatka earthquake. The examination focuses on the spatiotemporal distribution of magnetic data with Flag 8, which indicates discrepancies between the scalar and vector magnetometers. Flag 8 was observed along Swarm satellite orbits near the epicenter prior to the earthquake, initially appearing approximately 16 days before the mainshock and persisting until 9 days after. The results indicate that the spatial distribution of Flag 8 exhibits high density and extensive coverage around the epicentral region, showing a close relationship with the earthquake. Further examination suggests that neither polar geomagnetic activity nor the South Atlantic Anomaly can adequately explain the observed pattern. This implies that the observed Flag 8 is more likely associated with the seismic activity in the Kamchatka Peninsula. This study highlights Flag 8 as a complementary tool for detecting pre-earthquake magnetic anomalous phenomenon and advancing earthquake precursor research.

Keywords Ionosphere, seismo-ionospheric EM anomalies, Low-orbit satellite observation, Kamchatka Peninsula earthquake

1 Introduction

The United States Geological Survey (USGS) reported that a megathrust earthquake with a moment magnitude of 8.8 struck off the east coast of the Kamchatka Peninsula in the Russian Far East at 23:24:52 UTC on 29 July 2025. It is the largest event to occur worldwide since the 2011 M_w 9.1 Tohoku, Japan earthquake, ranking among the top ten largest earthquakes recorded since 1900 (Mori et al. 2011). According to the National Oceanic and Atmospheric Administration (NOAA), the mainshock generated a tsunami that propagated across the Pacific Ocean, which prompted tsunami alerts and resulted in damage and casualties. Volcanic activity around the Kamchatka Peninsula also intensified following the mainshock.

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Earthquakes are among the most destructive natural hazards threatening human societies (Ridzwan and Yusoff 2023; Shafapourtehrany et al. 2023). Research and development of pre-earthquake anomalous phenomena is one of the crucial strategies for earthquake hazard mitigation and risk reduction (Allen and Melgar 2019; Tritakis 2025; Kumar et al. 2022). In recent years, growing attention has been directed toward identifying potential pre-earthquake anomalous phenomena across various observable geophysical parameters (Bürgmann 2023; Bhardwaj et al. 2021), including electric fields (Ulu-kavak et al., 2016; Liu et al. 2013), ultra-low-frequency geomagnetic signals (Stănică et al., 2019), ionospheric disturbances (Chen et al. 2024; Liu et al. 2024), ionospheric plasma anomalies (Su et al. 2013), atmospheric temperature anomalies (Manesh et al. 2024), and radon emissions (Morales-Simfors et al., 2019). These observation results suggest that detectable disturbances occur in the lithosphere, atmosphere, and ionosphere prior to earthquakes, with possible interactions among them (Chen et al. 2023a, b). Given these multi-sphere manifestations, satellite-based measurements have emerged as a powerful tool for detecting pre-earthquake anomalous phenomena on a global scale, and increasingly enabled precise, global-scale characterization of pre-earthquake disturbances in the upper atmosphere and ionospheric parameters (Yang et al. 2023; Ouyang et al. 2018; Edvardsen et al. 2016). In particular, the Swarm constellation, which offers high spatial and temporal resolution for geomagnetic field monitoring, has been widely utilized for detecting pre-earthquake ionospheric anomalies potentially related to seismic activity (Qian et al. 2025; Xiong et al. 2021).

Numerous studies have reported the pre-earthquake ionospheric anomalies from Swarm data (Marchetti et al. 2020b; Harrigan et al., 2024; Balasis et al. 2024). For instance, Marchetti et al. (2019) identified magnetic and totalelectron density anomalous signals in four Swarm orbits before the 2016–2017 Amatrice-Norcia earthquakes, including a particularly strong magnetic anomalies 3 days prior to the mainshock. Satti et al. (2022) observed that 2 to 3 days before the mainshock, the Swarm satellites detected significant anomalies in multiple ionospheric parameters associated with a magnitude 6.5 earthquake, including variations in plasma density, slant total electron content and electron temperature. The reports of Balasis et al. (2024) show that the Swarm satellites detected ultra-low frequency pulsations and equatorial spread-F activity prior to the M6 earthquake in central Italy on August 24, 2016. These findings highlight the growing potential of Swarm satellite data in capturing pre-earthquake ionospheric anomalies.

The Swarm mission evaluates the quality of magnetometer data using a flagging system to categorize

observations. For example, Flag 0 signifies data collected under nominal conditions, while Flag 2 denotes the detection of outliers in the magnetic data, Flag 8 identifies discrepancies between Absolute Scalar Magnetometer (ASM) and Vector Field Magnetometer (VFM) measurements, and Flag 255 indicates VFM under-sampling (Billett et al. 2022). Typically, studies select data acquired under nominal conditions in VFM measurements (i.e., Flag 0) to ensure robust results (Marchetti et al. 2020a, b; Fournier et al. 2015). Data flagged for inconsistency between scalar and vector measurements (i.e., Flag 8) are generally excluded in previous studies. However, the inconsistency disappeared automatically without any article fix. This suggests that functions of magnetometer is normal. There is some factors dominate the inconsistency and worth to be investigated (De Michelis et al. 2017).

In this study, we focus on the Flag 8 occurrences from Swarm magnetic data prior to the 2025 M_w 8.8 Kamchatka earthquake. The relationship between spatio-temporal distribution of Flag 8 and the seismic event is examined. In addition, we discuss potential mechanisms that could produce Flag 8 such as polar geomagnetic activities and the South Atlantic Anomaly (SAA).

2 Data and methodology

Magnetic field observations from Swarm constellation were utilized as the data source. Swarm consists of three near-polar orbiting satellites (A, B, and C), launched on 22 November 2013 (Friis-Christensen, 2006). Two satellites fly in a lower orbit, while the third occupies a higher polar orbit, with orbital altitudes ranging from approximately 400 to 550 km (Fig. 1). The mission is designed to perform high-precision measurements of the geomagnetic field and its spatiotemporal variations. Each Swarm satellite carries a tri-axial fluxgate VFM, which operates in a 50-Hz sampling mode, together with an ASM that delivers highly accurate scalar measurements used for the in-flight calibration of the VFM.

Time-calibrated magnetic field data (SW_MAGx_LR_1B) at a 1-Hz sampling rate from the Level-1b magnetic products of the Swarm mission were utilized. The Level-1b products are calibrated, quality-screened time series and provide vector and scalar magnetic field time series, which have undergone instrument calibration, quality flagging, and unit conversion (available at <http://swarm-diss.eo.esa.int>).

The first step in processing Swarm magnetic data is to apply the mission's quality flags. The Swarm magnetic dataset provides the "Flags_B" field to indicate the quality of magnetic field measurements. These flags denote the respective operating states of the ASM and VFM under different conditions (Billett et al. 2022). Typically, only data with "Flag 0," which indicates the VFM is under

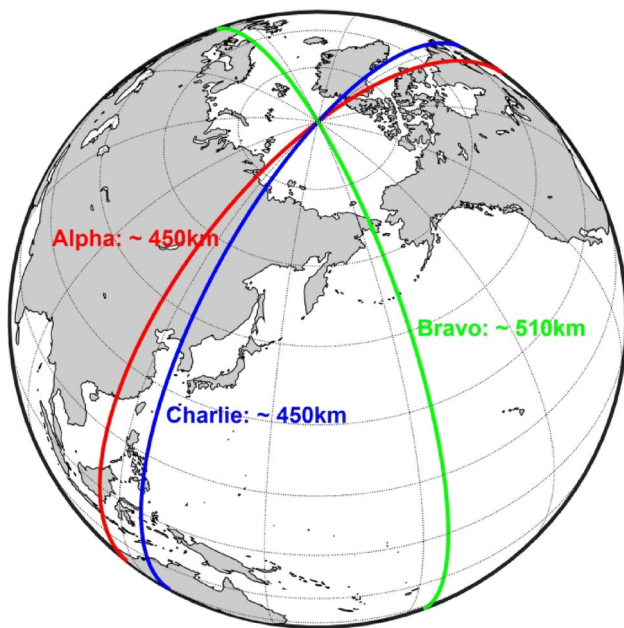


Fig. 1 Orbit of Swarm satellite. Swarm A (Alpha) and C (Charlie) form the lower pair of satellites flying side-by-side at an altitude of about 450 km, whereas Swarm B (Bravo) is cruising at a higher orbit of approximately 510 km

nominal conditions (Marchetti et al. 2020a; Fournier et al. 2015). In this study, we examined records flagged with a value of 8, which denotes notable discrepancies between the ASM and VFM measurements. We systematically investigated the temporal and spatial distribution of Flag 8 data and compared them with seismic and geomagnetic activities.

In addition to inspecting the Flag 8 events along individual satellite tracks, we summarized the spatiotemporal characteristics of Flag 8 using a gridded map. The map was divided into a grid system with a spatial resolution of $4^\circ \times 4^\circ$ in latitude and longitude, and for each cell, we calculated the frequency of samples labeled as Flag 8 within each 10-day period.

3 Results

3.1 Swarm observations of magnetic data with flag 8

We examined multiple Swarm orbital passes over the epicentral area during the month preceding the M_w 8.8 earthquake and observed that Flag 8 appeared near the epicenter, including two orbits on July 22 and 26 (Fig. 2a). During the July 22 orbit, Swarm-A traversed northward from the equator toward the pole, reaching a minimum distance of about 1,000 km from the epicenter. The magnetic field measurements from the ASM and VFM were consistent near the equator, and these measurements were marked as Flag 0 in the Swarm data quality reports. However, upon approaching the epicentral latitude,

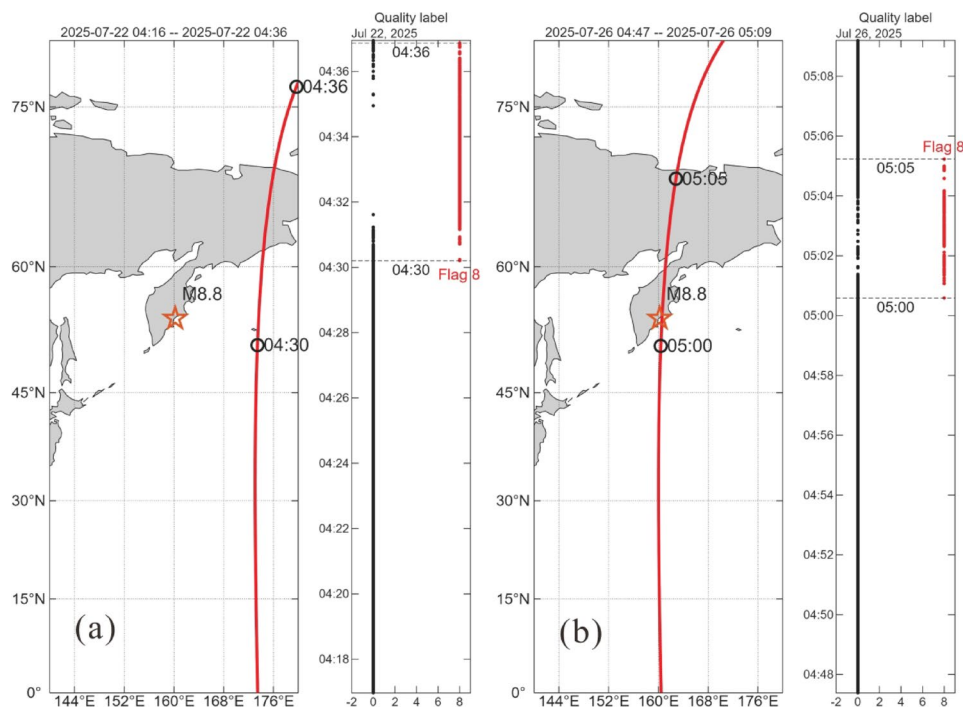


Fig. 2 Swarm satellite detected the Flag 8 in magnetic data as it approached the epicenter. (a) and (b) show two individual tracks of the swarm passing near the epicenter on July 16 and July 22, respectively. (The red line represents the satellite orbit, and black dots show the data under nominal condition, while the red dots indicate the discrepancies between the scalar and vector magnetometer records)

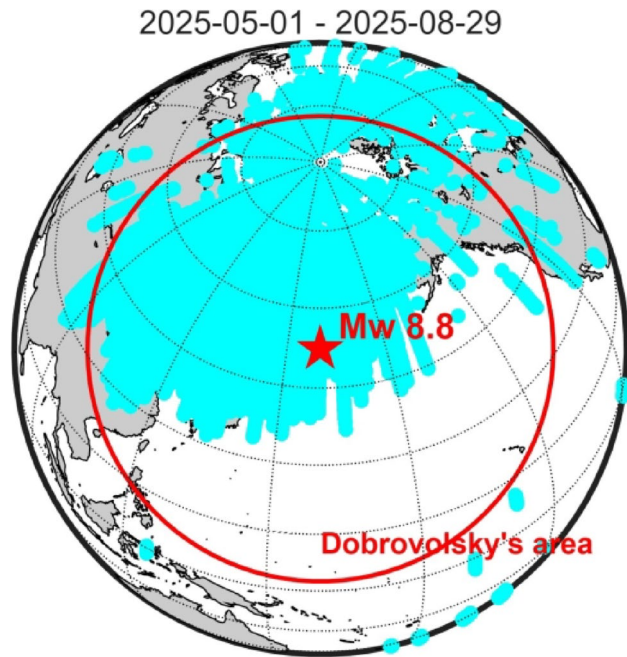


Fig. 3 Swarm magnetic data with Flag 8 from 31 May 2025 to 12 August 2025. The blue dots indicate the spatial locations of the Flag 8. The red circle denotes the earthquake preparation zone defined by Dobrovolsky's area

discrepancies between the ASM and VFM developed and were assigned the Swarm quality Flag 8. The Flag 8 in magnetic data were first recorded at 04:30 and disappeared by 04:36 as the spacecraft exited the affected region, lasting approximately 6 min.

On July 26, another satellite orbit recorded a similar phenomenon (Fig. 2b). The orbit crossed nearly directly over the epicenter, and the recorded Flag 8 in magnetic data appeared at 05:00 and disappeared at 05:05. At low latitudes, the magnetic measurements were marked

as Flag 0, however, Flag 8 appeared as the spacecraft neared the epicentral sector and subsided as the satellite departed. Combined with the observations on July 22, these results indicate that multiple Swarm orbits recorded a similar anomaly, the Flag 8 in magnetic data surged while passing over the epicenter, after which the VFM returned to nominal conditions after the satellite moved away, lasting about 4 to 6 min.

3.2 Spatiotemporal characteristics of flag 8 distribution

Magnetic data with Flag 8 were collected approximately three months prior to the earthquake. Figure 3 illustrates their spatial distribution from May 1 to August 14, 2025. The data with Flag 8 are densely concentrated and cover a large area near the epicenter, whereas only a few data are located in other regions. The spatial distribution of Flag 8 generally exhibits a semicircular pattern, skewed toward the northwest of the epicenter. To investigate the temporal distribution of this phenomenon, we computed the daily counts of Flag 8 points within the Dobrovolsky radius, 6,081 ($=10^{0.43 \times 8.8}$) km around the epicenter (Dobrovolsky et al. 1979).

The Fig. 4(a) shows the daily counts of Flag 8 in the Dobrovolsky's area (shown in Fig. 3), (b) demonstrate the quality flags of Swarm satellite magnetic data within 500 km of the epicenter, and (c) shows earthquakes ($M \geq 6$) occurred on the Kamchatka Peninsula. From May 31 to July 12, the selected region showed almost no Flag 8 in magnetic data. These Flag 8 data points emerged approximately 16 days before the earthquake, on July 14, and the daily counts increased gradually from around 200, reaching a maximum of over 1,600 on July 30. After the M_w 8.8 earthquake, Flag 8 data continued to be observed from August 1 to 9. The persistent observation of Flag 8 after the M_w 8.8 earthquake may be related to

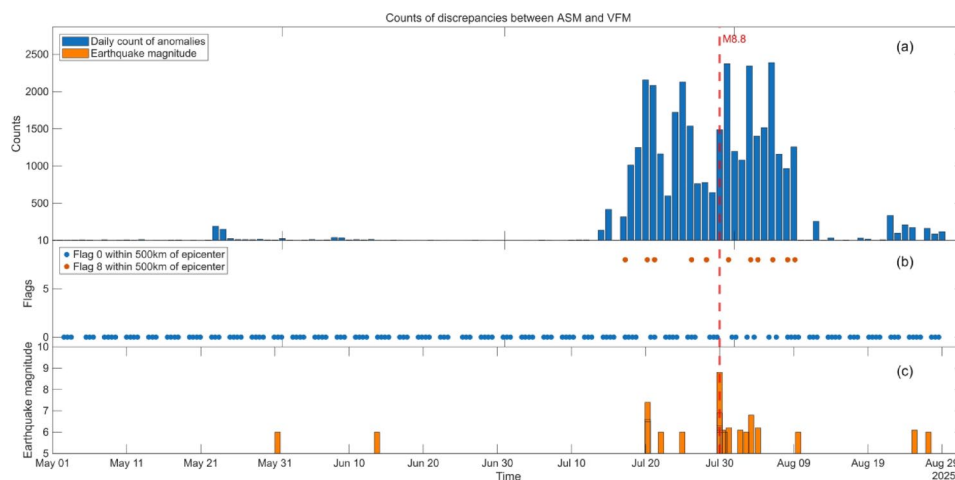


Fig. 4 The daily counts of magnetic data with Flag 8. (a) The blue bar chart indicates the number of Flag 8 in the Dobrovolsky's area (6,081 km around the epicenter). (b) The quality flags of Swarm satellite magnetic data within 500 km of the epicenter. Orange dots show the Flag 8 in magnetic data, while blue dots indicate the Flag 0. (c) The orange bar represents the magnitude of the earthquake ($M \geq 6$) that occurred only on the Kamchatka Peninsula

aftershocks in the Kamchatka Peninsula. The duration of this phenomenon was roughly 26 days in total, showing an initial increase followed by a gradual decrease in daily counts. Figure 4(b) present the Flag 8 occurrences within a 500 km radius of the epicenter. It can be observed that after narrowing the calculation range, Flag 8 appears almost exclusively around the time of the earthquake. During the month leading up to the earthquake and earlier, the Swarm magnetic field quality flags remained highly stable (Flag 0).

3.3 Temporal and spatial evolution of flag 8

The frequency of magnetic data with Flag 8 from June 30 to August 19 is shown in Fig. 5. During the interval from 30 to 20 days prior to the earthquake, magnetic data with Flag 8 were nearly absent near the epicenter. A local distribution of magnetic data with Flag 8 began to appear approximately on July 16. The data with Flag 8 surged and started to concentrate around the epicenter over the following days. Eight days later, the Flag 8 near the epicenter were densely distributed, covering a wide area from approximately 60°E to 180°E longitude and 30°N to 75°N latitude, from July 24 to August 3. The affected area spans roughly 5,000 km from north to south and about 8,000 km from east to west, with a total area of approximately $3.96 \times 10^7 \text{ km}^2$. This phenomenon persisted from August 1 to 11 following the M_w 8.8 earthquake. Following the earthquake, from August 9 to 19, it gradually disappeared.

4 Discussion

The coverage area of the observed Flag 8 is 31.6% larger than the earthquake preparation zone estimated by Dobrovolsky. This observation is consistent with the findings of Su et al. (2013), who reported that the ionospheric TEC anomalies prior to the M_w 7.1 Hector Mine earthquake surpassed the Dobrovolsky radius by approximately 30%.

The Flag 8 may result from multiple mechanisms that should be investigated (Michelis et al. 2017; De Michelis et al. 2017). The Auroral Electrojet (AE) index is widely used as an indicator of geomagnetic and electrojet activity in the auroral zone (Boudouridis et al. 2024; Zhong et al. 2021), and utilized to investigate the influence of polar geomagnetic activities. As a measure of polar geomagnetic activity, derived from data collected at high-latitude observatories, it provides a quantitative measure of the total deviation of the magnetic field from quiet-day values around the auroral oval (Qian et al. 2025; Singh et al. 2015).

AE index data are available at the World Data Center (WDC) for Geomagnetism, Kyoto, as shown in Fig. 6. During the period when Flag 8 was observed, from July 16 to August 9, 89.6% of the AE index values were below 500 nT, while only 0.4% exceeded 1000 nT. This indicates that polar geomagnetic activity was generally moderate to low and the geomagnetic environment was mostly stable during this period, with limited disturbances in the auroral electrojet. Although a weak geomagnetic storm might have occurred on August 10, the Flag 8 events that occurred had already ceased by that time. This observation indicates that the occurrences of Flag 8 in magnetic data prior to the M_w 8.8 earthquake are unlikely to be related to geomagnetic activity.

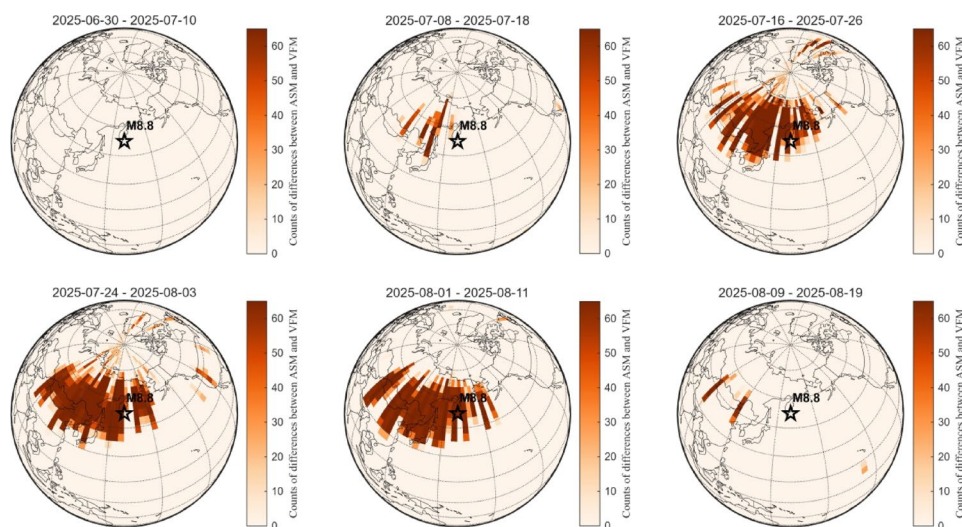


Fig. 5 The frequency of magnetic data with Flag 8 recorded by Swarm satellite. Colors indicating the number of data points with Flag 8 within each grid over a 10-day period

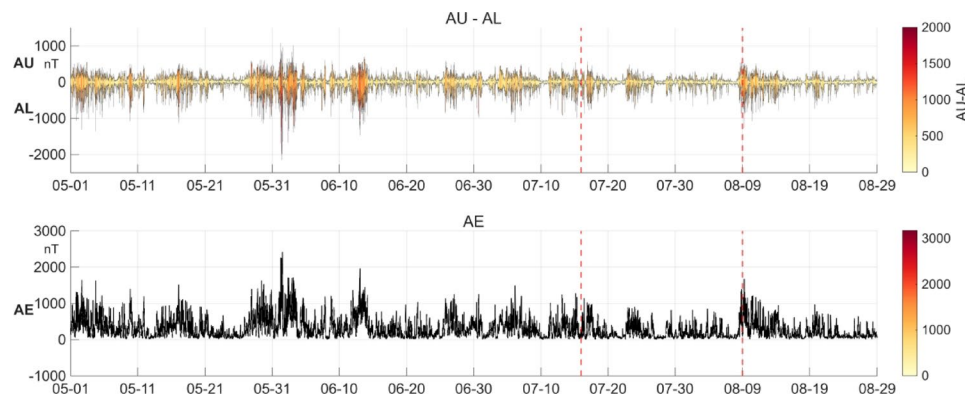


Fig. 6 The AE index in July 2025. The AU and AL index represent upper (AU) and lower (AL) envelopes of the superposed plots of all data from high-latitude observatories, red dotted line indicates the duration of the anomalies

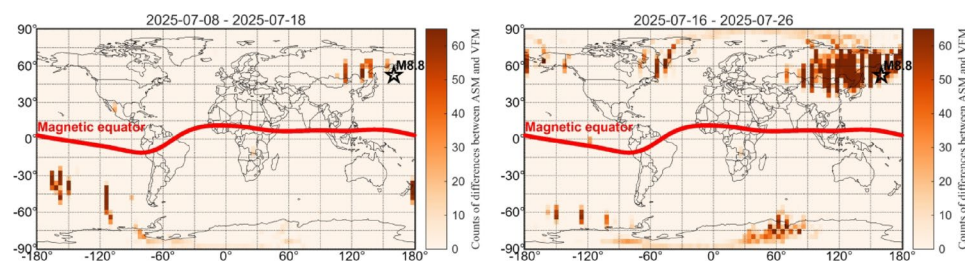


Fig. 7 Global distribution of Swarm magnetic data with Flag 8. The red line represents the magnetic equator. The distribution shows a significant concentration of Flag 8 around the Kamchatka earthquake epicenter, whereas the SAA region appears relatively quiet

We also considered the possible influence of the South Atlantic Anomaly (SAA). The Earth's magnetic field features a prominent weakened region over the South Atlantic Ocean, commonly referred to as the South Atlantic Anomaly (Kovář et al., 2021). The SAA is primarily caused by the non-concentric alignment between the Earth's geomagnetic dipole and its geographic center, which leads to a pronounced southward displacement of the magnetic equator over South America (Pavón-Carrasco et al., 2016). As a result, the magnetic field intensity in this region is significantly lower than at comparable latitudes elsewhere, making satellites more susceptible to energetic charged particles from the high-radiation zone at low altitudes (Ye et al. 2017).

However, the results show that almost no magnetic data with Flag 8 were observed within or near the SAA region during the period when they occurred near the epicenter (Fig. 7). This absence of Flag 8 data suggests that the observed phenomenon was not induced by the South Atlantic Anomaly or by energetic charged particles. The Flag 8 in magnetic data are more likely to originate from localized and transient phenomena, independent of the Earth's large-scale background magnetic features.

Figure 7. Global distribution of Swarm magnetic data with Flag 8. The red line represents the magnetic equator. The distribution shows a significant concentration of Flag 8 around the Kamchatka earthquake epicenter, whereas the SAA region appears relatively quiet.

We suggest that the observed Flag 8 likely originates from the transient response of the VFM sensor to external perturbations. According to Léger et al. (2015) and Fratter et al. (2016), the Swarm ASM package maintains sub-nT absolute accuracy and exceptional stability in both scalar and backup vector modes, serving as a reliable reference. While the VFM is rigorously calibrated using the scalar data, its fluxgate-based measurements are inherently more susceptible to rapid, direction-specific electromagnetic changes (Tøffner-Clausen et al. 2016). The similar phenomena can be consistently found in co-seismic magnetic perturbations that are obviously from fluxgate-based (VFM) measurements but not the proton (ASM) ones following seismic wave arrival (Aisa et al. 2025; Wang et al. 2019).

Chen et al. (2023a) proposed that the causal mechanism of double resonance in the seismo-lithosphere-atmosphere-ionosphere coupling before the earthquake. According to the causal mechanism, crustal resonances prior to an earthquake can trigger atmospheric resonance through the upward propagation of acoustic or internal gravity waves. This atmospheric resonance, in turn, could cause fluctuations in ionospheric plasma density (such as TEC anomalies) and disturbances in local ionospheric current systems (Chen et al. 2023b; Mao et al. 2025). Meanwhile, atmospheric resonance may somewhat resemble seismic waves in that it affects the magnetometers on the satellite, resulting in discrepancies between

VFM and ASM magnetic measurements. Although the underlying mechanism is not yet fully understood, we have attempted to investigate the relationship between Flag 8 and other earthquakes, using reliable and distinct data collected from other satellites to validate our suspecting.

5 Conclusions

Flag 8 in the Swarm magnetic data, which indicates discrepancies between the scalar and vector field magnetometers, was persistently observed near the epicenter in the month preceding the earthquake. The spatiotemporal characteristics of the magnetic data with Flag 8 were investigated and found to be strongly correlated with the M_w 8.8 earthquake. The magnetic data with Flag 8 densely concentrated around the epicenter, and appeared 16 days before the earthquake and gradually subsided within 9 days after the mainshock. We examined the potential mechanisms of the Flag 8, and the results reveal that polar geomagnetic activity and the South Atlantic Anomaly were unlikely to account for the observed pattern. Given their strong spatiotemporal correlation with the earthquake, these localized and short-lived Flag 8 events are most plausibly associated with seismic activity on the Kamchatka Peninsula.

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Author contributions

All authors reviewed the manuscript. Liwei Zhou: Writing – original draft, Formal analysis, Writing – review & editing; Shenjia Zhang: Investigation; Yinqian Li: Data curation, Software; Hong Liu: Software, Methodology; Tianya Luo: Data curation; Dan Tao: Investigation, Visualization; Yongxin Gao: Supervision; Longwei Chen: Resources; Jun Hu: Software, Validation; Xuemin Zhang: Resources, Methodology; Chieh-Hung Chen: Conceptualization, Supervision, Writing – review & editing;

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Data availability

All the data utilized in this paper is publicly available at <ftp://swarm-diss.esa.int>. The detailed information of flags in magnetic data could find at <https://swarmhandbook.earth.esa.int/catalogue/index>.

Declarations

Competing interests

The authors declare no competing interests.

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